WEIQI CHI

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EDUCATION

THE UNIVERSITY OF TOKYO (QS #39, 2027) TOKYO, JAPAN

Ph.D. student in Information Engineering Apr. 2024 – Mar.2027

** SPRING GX scholarship project student / Young Researcher's Encouragement Award, IEEE VTS 2024*

TOKYO INSTITUTE OF TECHNOLOGY (QS #97, 2027) TOKYO, JAPAN

Master's degree in Communication Engineering Oct. 2020 – Jun. 2022

** Supper Smart Society (SSS) scholarship project student*

RESEARCH

Learning-Based User Association & Mobility Management in mmWave V2X Networks Current

Doctoral Research

Investigated contextual multi-armed bandit (CMAB) approaches for user association (UA) in mmWave vehicular networks under non-stationary channel conditions without CSI acquisition or offline training:

- Developed BAND algorithm by integrating change-point detection (CUSUM-CD) with blockage-aware reward estimation to overcome the detection failure of conventional bandits under non-stationary channel reward brought by vehicle mobility and transient blockages. *(IEEE GLOBECOM, 2026)*
- Proposed a novel trajectory-aligned knowledge (TAK) regions in a semi-distributed CMAB framework for cross-vehicle knowledge sharing and algorithm cold-start acceleration with rigorous regret-bound analysis. *(IEEE TVT)*
- Proposed DCC-SMAB algorithm using dynamic hierarchical context clustering to mitigate slow convergence in high-dimensional context spaces for learning efficiency promotion. *(IEEE VNC, 2026)*
- Developed simulation framework integrating SUMO traffic simulator and ray-tracing tools with real-world maps for realistic V2X communication data analysis *(IEICE RCS, 2025)*.
- Proposed a geometric-based dynamic V2I blockage modeling and detection method. *(IEEE VTC, 2024)*

A Blockage-Aware mmWave V2X Connectivity Analysis based on MCM Nov. 2024 – Feb. 2025

Visiting Researcher at CARISSMA, Technische Hochschule Ingolstadt, Germany

- Collaborated with Artery core development team to design and implement ETSI ITS-G5 standards-compliant blockage prediction system with cooperative awareness on Artery simulation platform.
- Developed Maneuver Coordination Message (MCM) based V2X network simulation evaluated with OMNeT++.

Measurement and Prediction of Vegetation Attenuation on mmWave channel Sep. 2020 – Jul. 2022

Master Research & Joint Research with Rakuten Mobile, Inc.

- Designed and conducted outdoor radio measurement campaigns for static shadowing effects from vegetation to 5G base stations, validated results through ray-tracing simulation with propagation models (DKED, ITU-R).
- Presented research findings at Rakuten Mobile group discussions and student conferences (Best Oral Presenter).

FUNDINGS

ACT-X, Japan Science and Technology Agency (JST) Oct. 2026 – Mar. 2028

Passed first-round screening and interview.

ACT-X is one of the most prestigious grants available to PhD students in Japan, awarded by the country's leading national science funding body **JST** (comparable to NSF/DFG). Only 20 researchers within specified field were funded nationwide last year with direct funding of **¥6,000,000** (approx. **€33,000**) over 2 years.

PUBLICATION

Journal

[1] **Weiqi Chi**, B. Qian, L. Zhong, M. Kaneko, and M. Tsukada, "Demand-Aware Resource Allocation for Heterogeneous QoS Requirements in AI-Native 6G V2X Networks," IEEE Communications Magazine. *(Under submission, **JCR Q1・IF 8.3**)*

[2] **Weiqi Chi**, B. Qian, H. Wu, D. Li, Haibo Zhou, and M. Tsukada, "Toward Blockage-Resilient 6G-V2X Connectivity: Semi-Distributed Bandit with Dynamic Arm Set for mmWave HetNets," submitted to IEEE

Transactions on Vehicular Technology (**T-VT**), 2026. *(Under Review, **JCR Q1・IF 7.5+**)*

Internation conference

[1] S. Ma, B. Qian, Y. Nan, **Weiqi Chi**, T. Ma, and M. Tsukada, "UAV-LEO Collaborative Task Offloading in Edge Computing for Internet of Remote Things," in *Proc. 2026 Int. Conf. Meta-Networking (MEET)*, 2026. ***(Invited paper)***
[2] H. Wu, B. Qian, H. Si, **Weiqi Chi**, E. Javanmardi, Haibo Zhou, and M. Tsukada, " Sparsity and Evidential Uncertainty Enabled Communication and Sampling for Collaborative Semantic Occupancy Prediction," Proceedings of the AAAI Conference on Artificial Intelligence, 2027. *(Under Review, **CORE A***)*
[3] **Weiqi Chi**, and M. Tsukada, "Blockage-Aware Non-stationary Dynamic Bandit for User Association in mmWave V2X Networks," IEEE Global Communications Conference (**GLOBECOM**) 2026. ***(CORE B)***
[4] **Weiqi Chi**, X. He, and M. Tsukada, "Fast-Adapting V2X Networks: A Dynamic Context Clustering Multi-Armed Bandits," IEEE Vehicular Networking Conference (**VNC**), Canada, Jun. 2026.
[5] **Weiqi Chi**, J. Nakazato, T. Murakami and M. Tsukada, "V2I Blockage Modeling and Performance Evaluation for Connected Autonomous Vehicle", The IEEE 99th Vehicular Technology Conference (**VTC2024-Spring**), Singapore, Jun. 2024. ***(VTS Flagship)***
[6] **Weiqi Chi**, N. Keerativoranan, and J. Takada, "Outdoor mmWave Band Tx Localization Using Scattering Path under Spectrum Sharing Scenario," 13th Asia-Oceania Top University League on Engineering (**AOTULE**) Conference, Daejeon, Korea, Nov. 2021.

Domestic conference

[1] **Weiqi Chi**, J. Nakazato, T. Murakami, and M. Tsukada, "ミリ波帯における遮蔽対応型 V2I 通信と解決策について", in IEICE 無線通信システム研究会, Kyoto Institute of Technology, Mar. 2025.

ACADEMIC SERVICE

Reviewer for Journals

IEEE TCCN | *IEEE TNSE* | *IEEE Network Magazine*

Technical Program Committee Member

International Conference on Aerospace Information Perception and Intelligent Processing (AIPIP 2026)

IEEE INFOCOM 2026, Tokyo, Japan May. 2026

Student Volunteer | *Served as cameraman and logistics support staff for 900+ attendees from 40 countries*

INTERNSHIP

Visban Inc., Tokyo, Japan Jun. 2026 – Current

Intern | *mmWave repeater simulation engineer* | Sionna Raytracing, Network Controlled Repeater

Tier IV Inc., Tokyo, Japan Apr. 2026 – Sep. 2026

Intern | *Autoware V2X engineer* | Autoware, Veins, QuaDRiGa

Ehime Prefectural Industrial Technology Institute, Japan Aug. 2025

Intern | *AR/VR Application Development* | MRTK, Unity, C#, HoloLens2

General Test Systems Inc., Shenzhen, China Feb. 2023 – Dec. 2023

Intern | *Application engineer* | OTA testing, Anechoic chamber, VNA, Area Tester

Rakuten Mobile Inc. R&D group, drone application development, Japan Sep. 2022

Intern | *Drone application development* | DJI drones operation, SDK, Android platform

SKILL

- **Programming skills:** C, C++, MATLAB, NED language
- **Development & Tools:** Git, LaTeX, Linux/Ubuntu
- **Hardware:** Vector Network Analyzer (VNA), Spectrum Analyzer, OTA testing
- **Simulation & Modeling:** OMNeT++, SUMO, Artery, Veins, Wireless InSite, Sionna
- **Language:** Chinese (Native), English (Native level) , Japanese (JLPT N1) , German (Beginner)
- **Interests:** Piano, Music, Swimming, Boxing, Bouldering